

Markscheme

Mathematics: analysis and approaches

Practice question bank

15. (a)

Consider the function $f(x) = \frac{2x + 13}{(x - 1)(x + 4)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x - 1} + \frac{B}{x + 4}$.

$$2x + 13 = A(x + 4) + B(x - 1) \quad (M1)$$

$$\text{substituting } x = 1: 15 = 5A \Rightarrow A = 3 \quad \mathbf{A1}$$

$$\text{substituting } x = -4: 5 = -5B \Rightarrow B = -1 \quad \mathbf{A1}$$

[3 marks]